Practice Assignment 3 - Not Graded

This assignment will not be graded and is only for practice.

Multiple Select Questions (MSQ)

1 point

Which of the following sets with the given addition and scalar multiplication (scalars are real numbers in every case) form vector spaces?

$V_1 = \{(x, y, z) \mid x, y, z \in R\}$
Addition: $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$; $(x_1, y_1, z_1), (x_2, y_2, z_2) \in V_1$ V1 =
Scalar multiplication:

{(x, y, z) | x, y, z ∈ R}Addition: (x1, y1, z1) + (x2, y2, z2) = (x1 + x2, y1 + y2, z1 + z2);
(x1, y1, z1), (x2, y2, z2) ∈ V1Scalar multiplication:

$c(x, y, z) = \begin{cases} (0, 0, 0) & c = 0 \\ (cx, cy, z) & c \neq 0 \end{cases} \quad (x, y, z) \in V_1,\ c \in R$c(x, y, z) = {(0, 0, 0)(cx, cy, cz)
c = 0c = 0 (x, y, z) ∈ V1, c ∈ R

$V_2 = \{(x, y, z) \mid x, y, z \in R\}$
Addition: $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 z_2)$; $(x_1, y_1, z_1), (x_2, y_2, z_2) \in V_2$ V2 =
Scalar multiplication: $c(x, y, z) = (cx, cy, cz)$; $(x, y, z) \in V_2,\ c \in R$

{(x, y, z) | x, y, z ∈ R}Addition: (x1, y1, z1) + (x2, y2, z2) = (x1 + x2, y1 + y2, z1z2);
(x1, y1, z1), (x2, y2, z2) ∈ V2Scalar multiplication: c(x, y, z) =
(cx, cy, cz); (x, y, z) ∈ V2, c ∈ R

$V_3 = \{(x, y, z) \mid x, y, z \in R\}$
Addition: $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$; $(x_1, y_1, z_1), (x_2, y_2, z_2) \in V_3$ V3 =
Scalar multiplication: $c(x, y, z) = (cx, cy, z)$; $(x, y, z) \in V_3,\ c \in R$

{(x, y, z) | x, y, z ∈ R}Addition: (x1, y1, z1) + (x2, y2, z2) = (x1 + x2, y1 + y2, z1 + z2);
(x1, y1, z1), (x2, y2, z2) ∈ V3Scalar multiplication: c(x, y, z) =
(cx, cy, z); (x, y, z) ∈ V3, c ∈ R

$V_4 = \{(x, y, z) \mid x, y, z \in R, x + y + z = 1\}$
Addition: $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$; $(x_1, y_1, z_1), (x_2, y_2, z_2) \in V_4$ V4 =
Scalar multiplication: $c(x, y, z) = (cx, cy, cz)$; $(x, y, z) \in V_4,\ c \in R$

{(x, y, z) | x, y, z ∈ R, x + y + z = 1}Addition: (x1, y1, z1) + (x2, y2, z2) = (x1 + x2, y1 + y2, z1 + z2);
(x1, y1, z1), (x2, y2, z2) ∈ V4Scalar multiplication: c(x, y, z) =
(cx, cy, cz); (x, y, z) ∈ V4, c ∈ R

$V_5 = \{(x, y) \mid x, y \in R, x + 2y = 0\}$
Addition: $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$; $(x_1, y_1), (x_2, y_2) \in V_5$ V5 =
Scalar multiplication: $c(x, y) = (cx, cy)$; $(x, y) \in V_5,\ c \in R$

{(x, y) | x, y ∈ R, x + 2y = 0}Addition: (x1, y1) + (x2, y2) = (x1 + x2, y1 + y2); (x1, y1), (x2, y2) ∈ V5Scalar multiplication: c(x, y) = (cx, cy); (x, y) ∈ V5, c ∈ R

[Hint:Hint: Check the axioms related to scalar multiplication in V_1 V1.]

- [ ] V_1 V1 is a vector space.
- [ ] V_2 V2 is a vector space.
- [ ] V_3 V3 is not a vector space.
- [ ] V_4 V4 is a vector space.
- [ ] V_5 V5 is a vector space.

No, the answer is incorrect.
Score: 0
Accepted Answers:
V_3 V3 is not a vector space.
V_5 V5 is a vector space.

1 point

The marks obtained by Karthika, Romy and Farzana in Quiz 1, Quiz 2 and End sem (with the maximum marks for each exam being 100) are shown in Table M2W3P1. Let Quiz 1, Quiz 2 and End sem marks obatined by Karthika, Romy and Farzana be represented by vectors. Use the above information, to choose the correct option(s).

- [ ] If Farzana obtained 90 marks in End sem (i.e $c = 90$c = 90), then the marks in Quiz 1, Quiz 2 and End sem represent linearly independent vectors.
- [ ] If Farzana obtained 80 marks in End sem (i.e $c = 80$c = 80), then the marks in Quiz 1, Quiz 2 and End sem represent linearly independent vectors.
- [ ] If 20% of Quiz 1, 30% of Quiz 2, and 50% of End sem are used to obtain total marks, then the vector representing the total marks is a linear combination of vectors representing the marks of Quiz 1, Quiz 2 and End sem.
- [ ] If 0% of Quiz 1, 50% of Quiz 2 and 50% of End sem are used to obtain total marks, then the vector representing the total marks is not a linear combination of vectors representing the marks of Quiz 1, Quiz 2 and End sem.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If Farzana obtained 80 marks in End sem (i.e $c = 80$c = 80), then the marks in Quiz 1, Quiz 2 and End sem represent linearly independent vectors.
If 20% of Quiz 1, 30% of Quiz 2, and 50% of End sem are used to obtain total marks, then the vector representing the total marks is a linear combination of vectors representing the marks of Quiz 1, Quiz 2 and End sem.

1 point

Choose the set of correct options.

- [ ] Any subset of any linearly independent set of vectors is linearly independent.
- [ ] Any subset of any linearly dependent set of vectors is linearly dependent.
- [ ] Any subset of any linearly dependent set of vectors is linearly independent.
- [ ] Any set of 3 vectors in R^3 is linearly independent.
- [ ] Any set of 4 vectors in R^3 is linearly dependent.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Any subset of any linearly independent set of vectors is linearly independent.
Any set of 4 vectors in R^3 is linearly dependent.

Numerical Answer Type (NAT)

Consider the set $V = \{(-1, x, -y) \mid x, y \in R\} \subseteq R^3$ with the usual addition and scalar multiplication as in R^3 and associated statements given below.

- P: V is not closed under addition.
- Q: V is not closed under scalar multiplication.
- R: V has a zero element with respect to addition.i.e., there exists some element w such that $v + w = v$, for all $v \in V$.
- S: V is a vector space.

Find out the number of true statements.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point

Consider the set of three vectors $S = \{(1,18,16),(1,c-2,16),(1,18,c-4)\}$ in R^3 with usual addition and scalar multiplication. Find the value of c, so that the above set S will be linearly dependent?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 20
1 point

For how many values of a is the set $\{(1,2,a),(a,0,a,),(-1,a^2,8),(0,1,-3a)\}$ linearly independent?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0

1 point

Comprehension Type Question:

Suppose in a village there are four farmers A, B, C and D, each owning 1 acre of land. They cultivate paddy, pulses and/or sugarcane in their lands as follows: Farmer A uses 50% of his land for paddy, 30% for pulses and the remaining for sugarcane. Farmer B uses 40% of her land for paddy and she divides her remaining land equally for pulses and sugarcane. Farmer C uses the whole land for paddy only, and Farmer D uses 30% for paddy, 30% for pulses and the remaining for sugarcane. Using the above information to answer the following questions.

1 point

Suppose the area used by a farmer for different crops is denoted by a row vector. Let S be the span of the resulting four row vectors . Choose the correct set of options.

- [ ] The row vectors corresponding to the area used for different crops by Farmers A, B and D are linearly dependent.
- [ ] The row vectors corresponding to the area used for different crops by Farmers A, B, C and D are linearly dependent.
- [ ] The row vectors corresponding to the area used for different crops by Farmers A, B and C are linearly dependent.
- [ ] The row vectors corresponding to the area used for different crops by Farmers A, C and D are linearly dependent.

No, the answer is incorrect.
Score: 0
Accepted Answers:

The row vectors corresponding to the area used for different crops by Farmers A, B and D are linearly dependent.
The row vectors corresponding to the area used for different crops by Farmers A, B, C and D are linearly dependent.

Suppose Farmer B buys the same amount of land (1 acre) and uses it in the same ratio for different crops as she was using for her land earlier. Consider the following statements.

Statement 1:Statement 1: Farmer A sells his whole land to Farmer D, and Farmer D uses the land she bought from Farmer A, in the same ratio for different crops as Farmer A was using earlier. In this new scenario, Farmer B is using more amount of area of land for pulses than Farmer D.

Statement 2:Statement 2: Farmer A sells his whole land to Farmer D, and Farmer D uses the land she bought from Farmer A, in the same ratio for different crops as Farmer A was using earlier. In this new scenario, Farmer B and Farmer D are using the same amount of area of land for paddy.

Statement 3:Statement 3: Farmer A sells his whole land to Farmer C, and Farmer C uses the land she bought from Farmer A, in the same ratio for different crops as Farmer A was using earlier. In this new scenario, Farmer B and Farmer C are using the same amount of area of land for paddy.

Statement 4:Statement 4: Farmer A sells his whole land to Farmer C, and Farmer C uses the land she bought from Farmer A, in the same ratio for different crops as Farmer A was using earlier. In this new scenario, Farmer B is using more amount of area of land for paddy than Farmer C.

Write down the number corresponding to the correct statement. (If Statement 2 is right, the answer would be 2.)

No, the answer is incorrect.
Score: 0
Accepted Answers:

(Type: Numeric) 2
1 point